MG4DSEHISA00: SCIENCE AND MODERNITY

Answer Scheme

Part A

1. Laws of Motion and Universal Gravitation- Development of Scientific Method and Mathematical tools

2. Analytical geometry (also called coordinate geometry) is a branch of mathematics that studies geometric shapes and figures using algebraic equations and a coordinate system (usually the Cartesian plane)- It connects algebra and geometry by representing points, lines, and curves with numerical values and equations.

3. Descartes proposed systematic skepticism—doubting everything that can possibly be doubted (including the senses and external world) to find an absolutely certain foundation of knowledge. This led to the famous conclusion "Cogito, ergo sum" ("I think, therefore I am"), which he considered indubitable truth- The method of doubt laid the foundation for modern philosophy by emphasizing reason and critical inquiry.

4. Earth-centered universe: The Earth is stationary and lies at the center of the universe, while the Sun, Moon, planets, and stars revolve around it in circular paths- Use of epicycles and deferents: To explain irregular planetary motions (like retrograde motion), planets move in small circles (epicycles) whose centers move along larger circles (deferents) around the Earth.

5. Exchange of ideas and practices: It involves the transfer of knowledge, beliefs, languages, art, and traditions between societies through contact such as trade, migration, or communication- Mutual influence and transformation: Different cultures influence each other, leading to changes, adaptation, and the creation of new cultural forms.

6. Gendered Targeting of Women: Witch-hunts disproportionately targeted women, especially those who were socially marginal (widows, elderly, or independent women). Cultural beliefs associated women with emotional instability, sexuality, and moral weakness, making them more vulnerable to accusations of witchcraft- Reinforcement of Patriarchal Control:Witch-hunts functioned as a mechanism to discipline and control women who deviated from accepted gender roles. By punishing "deviant" behavior, societies reinforced patriarchal norms and limited women's autonomy and social power.

7. Universe as a Machine: The idea of the Mechanical Universe views nature as a vast, orderly machine governed by fixed laws of motion, especially associated with thinkers like Isaac Newton. All physical phenomena are explained through matter in motion and predictable cause-and-effect relationships- Determinism and Predictability:It assumes that if the position and motion of all objects are known, the future state of the universe can be precisely predicted. This reflects a deterministic worldview.

8. Market-oriented production: Commercial agriculture refers to farming where crops and livestock are produced primarily for sale in the market rather than for subsistence, often involving large-scale operations- Use of modern inputs and technology: It typically relies on advanced techniques such as machinery, irrigation, chemical fertilizers, and improved seeds to increase productivity and profit.

9. Scientific temper refers to an attitude of logical thinking, questioning, and reliance on evidence rather than superstition. It is enshrined as a Fundamental Duty under Article 51A(h) of the Constitution of India, which urges citizens to develop a scientific temper, humanism, and a spirit of inquiry and reform- It promotes rationality, critical thinking, and openness to new ideas, helping society progress, combat superstition, and support democratic and social development.

10. Small-scale, individual research: "Little Science" refers to early forms of scientific activity carried out by individual scientists or small groups, often with limited resources and simple equipment- Pre-institutional and less organized: It existed before the rise of large research institutions and state-funded projects, relying more on personal initiative than on formal organization or collaboration.

11. Unity of Science: Science is unified by shared methods, principles, and goals—such as empirical observation, experimentation, and the search for universal laws that explain natural phenomena- Differentiation within Science: At the same time, science is differentiated into specialized disciplines (e.g., physics, biology, chemistry), each focusing on specific domains, methods, and levels of analysis, reflecting the complexity and diversity of knowledge.

12. Collective Action with Shared Goals-A social movement involves organized efforts by a group of people who share common interests, aims, or grievances and work together to bring about or resist social change- Sustained and Organized Effort- It is not a one-time activity but a continuous, structured process that uses various strategies (like protests, campaigns, or advocacy) to influence society or policy over time.

Part B

13. Global Knowledge Networks-Jesuit scholars created extensive international networks (Europe, Asia, Americas) that facilitated the exchange of scientific knowledge, especially in astronomy, geography, and natural history-Advancement of Astronomy and Mathematics-Figures like Christopher Clavius contributed to calendar reform (Gregorian calendar) and promoted mathematical sciences within education and research-Science through Missionary Work-Jesuits such as Matteo Ricci introduced European scientific ideas (astronomy, cartography) to China while also bringing back indigenous knowledge to Europe-Educational Institutions and Scientific Training-Jesuits established colleges and universities across Europe that emphasized rigorous training in mathematics, natural philosophy, and experimental methods-Mediation between Science and Religion-Jesuit scholars played a key role in reconciling scientific discoveries with Catholic doctrine, engaging with debates around heliocentrism and natural philosophy during the early modern period.

14. Institutional platforms for knowledge exchange-Scientific societies like the Royal Society created formal spaces for scholars to meet, debate, and present findings, helping transform science into a collective and organized activity-Dissemination through journals-Journals such as the Philosophical Transactions enabled rapid circulation of new discoveries across regions, making scientific knowledge more accessible and less dependent on personal networks-Standardization and validation of knowledge-These institutions introduced practices like peer evaluation and replication, which helped establish credibility, accuracy, and uniform methods in knowledge production-Creation of scientific communities and networks-They connected scientists across countries, fostering collaboration, cross-cultural exchange, and the emergence of an international scientific community-Transformation of knowledge production-Scientific societies and journals shifted knowledge production from isolated individuals to a collective, cumulative, and institutional process, accelerating innovation and laying the foundation for modern "Big Science."

15. Air and Environmental Pollution-Rapid industrialization led to heavy smoke, coal dust, and toxic emissions, causing respiratory diseases such as bronchitis and asthma-Poor Working Conditions-Factory workers were exposed to dangerous machinery, long hours, and unsafe environments, resulting in frequent injuries, accidents, and occupational illnesses-Urban Overcrowding and Sanitation Issues-Mass migration to industrial cities created slums with inadequate housing, poor drainage, and lack of clean water, leading to diseases like cholera and typhoid-Child and Women Labour Exploitation-Children and women worked in hazardous conditions with little protection, causing physical deformities, fatigue, and long-term health problems-Spread of Epidemics and Public Health Crises-Unsanitary living and working conditions contributed to frequent outbreaks of infectious diseases, increasing mortality rates in industrial towns.

16. Father of Modern Philosophy-Descartes emphasized reason and doubt as the basis of knowledge, marking a shift from medieval scholasticism to modern philosophy-Method of Doubt-He introduced systematic skepticism—questioning all beliefs to arrive at certain truth, famously expressed as "Cogito, ergo sum" ("I think, therefore I am")-Development of Analytical Geometry-Descartes linked algebra and geometry, creating the Cartesian coordinate system, which became fundamental to modern mathematics-Mind–Body Dualism-He proposed that mind and body are distinct substances, influencing later debates in philosophy, psychology, and science-Contribution to Scientific Method-Descartes advocated a rational, mathematical approach to science, helping shape the development of modern scientific thinking.

17. Classification and Census Practices-Governments used racial science to classify populations into rigid racial categories through censuses, reinforcing hierarchies and fixed identities-Segregation Policies-Administrative systems enforced racial segregation in housing, education, and public services, often justified as "scientifically" necessary-Immigration Control-Racial theories shaped restrictive immigration laws, favoring certain races while excluding others deemed "inferior."-Employment and Labor Regulation-Policies allocated jobs and wages based on racial classifications, creating systemic inequalities in labor markets-Legal and Political Discrimination-Racial science informed laws that limited political rights (like voting) and legal protections for marginalized racial groups.

18. Power is Diffuse and Everywhere-Power is not concentrated in a single authority (like the state) but is spread across society and exercised through everyday relationships and practices-Power and Knowledge are Interlinked-Power produces knowledge and vice versa ("power/knowledge"). What counts as truth is shaped by power structures, such as institutions and discourses-Power is Productive, Not Just Repressive-Power does not only repress or control; it also creates realities, identities, norms, and forms of behavior (e.g., shaping what is considered "normal")-Power Operates through Discourses-Systems of thought (discourses) define and regulate knowledge, influencing how people think, speak, and understand the world-Power is Relational and Dynamic-Power exists in relationships and is constantly shifting; where there is power, there is also resistance, making it fluid rather than fixed.

19.Eurocentrism as Ideology-Amin argues that Eurocentrism is not neutral knowledge but an ideological construct that presents Europe as the center and driver of world history-False Universalism-It falsely universalizes European historical experiences (like capitalism and modernity) as a model for all societies, ignoring diverse developmental paths-Distortion of Global History-Eurocentrism marginalizes and undervalues contributions of non-European civilizations, especially in Asia and Africa, to science, culture, and economy-Justification of Capitalist Domination-Amin links Eurocentrism to global capitalism, arguing it legitimizes imperialism, colonialism, and unequal development between the "core" and "periphery."-Call for Delinking and Plurality-He advocates "delinking" from Eurocentric models and emphasizes the need for alternative, plural paths of development rooted in local histories and conditions.

20. Colonial encounters and knowledge exchange-Hybrid scientific practices emerged in colonial contexts where European science interacted with indigenous knowledge systems. Colonial administrations depended on local expertise (e.g., in medicine, agriculture), leading to blended forms of knowledge rather than pure imposition-Power, hierarchy, and epistemic dominance-Scientific hybridity was shaped by unequal power relations—Western science was often privileged as "universal," while local knowledge was subordinated, yet still selectively incorporated, producing hybrid forms under asymmetrical authority-Role of intermediaries and knowledge brokers-Translators, native informants, missionaries, and local elites mediated between knowledge systems. These actors played a crucial role in adapting, translating, and reshaping scientific ideas across cultural boundaries-Institutional settings and state interests-Colonial institutions (botanical gardens, surveys, medical colleges) encouraged hybridization to serve administrative needs like resource extraction, governance, and public health, blending empirical local practices with European frameworks-Resistance, adaptation, and localization-Indigenous communities and practitioners did not passively accept Western science; they adapted, resisted, and reinterpreted it. This resulted in localized, hybrid scientific practices reflecting both global influences and local priorities.

21. Systematic Knowledge-Science is an organized and systematic way of understanding the natural and social world through careful observation and study-Based on Evidence-It relies on empirical evidence—facts gathered through experiments, observation, and measurement—rather than beliefs or opinions-Use of Scientific Method-Science follows a structured process (question, hypothesis, experimentation, analysis, conclusion) to arrive at reliable knowledge-Objective and Rational-It aims to be objective, logical, and free from personal bias, using reasoning and critical

thinking to explain phenomena-Dynamic and Evolving-Scientific knowledge is not fixed; it continuously changes and improves as new discoveries and better explanations emerge.

22. Division of Knowledge-As scientific knowledge expanded, it became too vast for one person to master, leading to specialization into narrower areas of study- Depth over Breadth:Specialization allowed scientists to focus deeply on specific problems, resulting in the development of distinct disciplines (e.g., physics, chemistry, biology)-Emergence of Sub-disciplines: Within existing sciences, further specialization created new fields (e.g., molecular biology, astrophysics) based on refined methods and topics-Technological Advancement: New instruments and techniques (like microscopes, particle accelerators) required specialized skills, encouraging the formation of new scientific areas-Institutional Support: Universities, research institutes, and journals began organizing around specialized fields, reinforcing their identity and promoting further growth.

Part C

23. Impact of Galileo's Experimental–Mathematical Approach & Its Long-Term Influence

Fusion of Experiment and Mathematics-Galileo Galilei combined systematic experimentation with mathematical formulation, establishing a new method for studying nature-Shift from Qualitative to Quantitative Science-Replaced Aristotelian qualitative explanations with measurable, numerical analysis of physical phenomena-Foundation of Modern Physics-His work on motion (falling bodies, inertia) laid the groundwork for classical mechanics-Validation of Controlled Experimentation-Emphasized repeatable experiments (e.g., inclined plane studies), making empirical verification central to science-Challenge to Authority and Tradition-Undermined reliance on ancient authorities like Aristotle, promoting independent inquiry-Support for Heliocentrism-Provided observational evidence for Nicolaus Copernicus's theory, strengthening the scientific revolution-Influence on Newtonian Synthesis-Inspired Isaac Newton, whose laws of motion and gravitation built directly on Galileo's mathematical treatment of motion-Standardization of Scientific Method-Helped shape the modern scientific method: hypothesis → experiment → mathematical analysis → conclusion-Technological and Instrumental Advancement-Improved use of instruments like the telescope, encouraging precision and innovation in scientific tools-Long-term Legacy Across Sciences-His approach influenced later scientists across disciplines, embedding mathematization and experimentation as core principles of modern science.

24.Emphasis on Reason over Tradition-Rationalisation prioritizes logic and reason instead of religious beliefs or traditional authority, forming the basis of modern scientific inquiry-Systematic Methodology-It promotes structured methods like observation, experimentation, and verification, leading to standardized scientific practices-Quantification and Measurement-Reality is increasingly understood through numbers, measurement, and mathematical models, enhancing precision and objectivity-Objectivity and Neutrality-Rationalisation encourages detachment from personal biases, aiming for universal and objective scientific truths-Disenchantment of Nature-Nature is no longer seen as mystical but as governed by laws that can be discovered and explained scientifically-Development of Scientific Laws-It enables the formulation of general laws (e.g., laws of motion, thermodynamics) that explain natural phenomena predictably-

Specialisation and Differentiation-Rationalisation leads to the division of science into specialized disciplines, increasing depth and expertise-Institutionalisation of Science-Scientific research becomes organized within universities, laboratories, and professional bodies, ensuring systematic knowledge production-Technological Advancement-Rational thinking drives innovation, linking science with technology and industrial development-Bureaucratisation of Knowledge-Scientific work becomes regulated, documented, and standardized, often guided by formal procedures and peer review systems.

25.Meaning of Vernacularisation-Refers to the translation and adaptation of scientific knowledge from European languages into local (vernacular) languages in colonial societies-Colonial Context-Initiated under colonial rule to make Western science accessible to local populations, often for administrative, educational, and missionary purposes-Role of Translation-Scientific texts were translated into regional languages (e.g., Bengali, Tamil, Hindi), enabling wider dissemination beyond elite, English-educated groups-Mediating Knowledge Systems-Vernacularisation acted as a bridge between Western scientific ideas and indigenous knowledge traditions, sometimes blending the two-Educational Reforms-Colonial governments and missionaries introduced vernacular science education in schools to expand literacy and technical skills among the masses-Print Culture and Publishing-Growth of printing presses and vernacular journals helped circulate scientific ideas more widely in colonial societies-Creation of New Terminology-Local languages developed new scientific vocabularies, often by borrowing, adapting, or inventing terms to express modern concepts-Agency of Indigenous Intellectuals-Local scholars, teachers, and reformers actively participated in translating, interpreting, and reshaping scientific knowledge-Selective Adaptation and Resistance-Not all Western science was accepted wholesale; local communities sometimes resisted or reinterpreted ideas based on cultural beliefs-Long-term Impact-Contributed to the democratization of knowledge, growth of scientific awareness, and the eventual rise of nationalist movements that linked science with progress and self-reliance.

26.Rise of Specialisation-Knowledge expanded rapidly, leading to division into distinct fields (e.g., physics, chemistry, biology), replacing the earlier unified “natural philosophy.”-Institutionalisation of Science-Universities and research institutes formalised disciplines through dedicated departments, chairs, and laboratories-Professionalisation of Scientists-Science became a full-time career with training, qualifications, and recognition, replacing amateur naturalists-Development of Scientific Societies-Organisations like the Royal Society promoted disciplinary boundaries through meetings, funding, and collaboration-Growth of Scientific Journals-Specialised journals emerged, allowing discipline-specific communication and standardisation of knowledge-Impact of Industrial Revolution-Industrial needs (engineering, chemistry, metallurgy) pushed applied sciences to develop as independent disciplines-Laboratory-based Research-The shift to experimental, lab-based science (e.g., in chemistry and physics) strengthened disciplinary identity-Standardisation of Methods and Terminology-Each discipline developed its own methods, tools, and vocabulary, distinguishing it from others-State and Patronage Support-Governments funded scientific research for military, economic, and administrative purposes, reinforcing specialised fields-Epistemological Differentiation-Thinkers like Auguste Comte classified sciences hierarchically, legitimising their separation into distinct disciplines.